

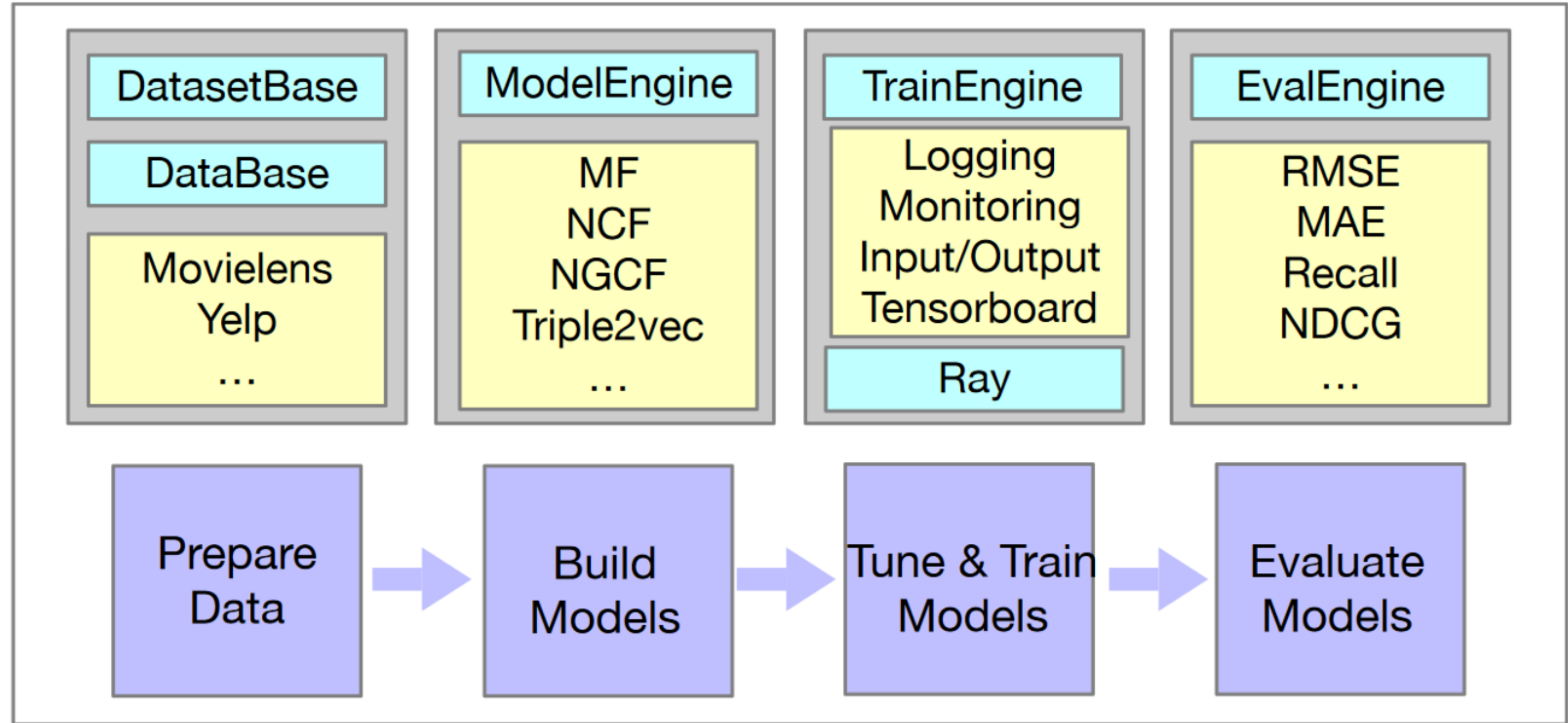
Overview

- Beta-RecSys is an **open source** project for Building, Evaluating and Tuning Automated Recommender Systems. It provides:
 - Built-in means for dataset preparation and splitting using common strategies
 - A generalized model engine for implementing **neural recommender models** using Pytorch along with a wide range of available out-of-the-box models
 - A unified training, validation, tuning and testing pipeline.
 - BetaRecSys is designed to be both **modular** and **extensible**, enabling new models to be quickly added to the framework.

Key Features

- Contains a convenient and reusable **dataset preparing toolkit** for processing raw datasets in a standardized manner.
 - Provides a **unified framework** for training models, monitoring the training processes, as well as validation and testing of the resulting models.
 - Provides 15+ **classical and neural recommendation models** that can be used out-of-the-box.
 - Supports **containerized deployment** for use in different environments.
 - Integrates the **Ray hyperparameter tuning library**
 - <https://github.com/ray-project/ray>

Architecture



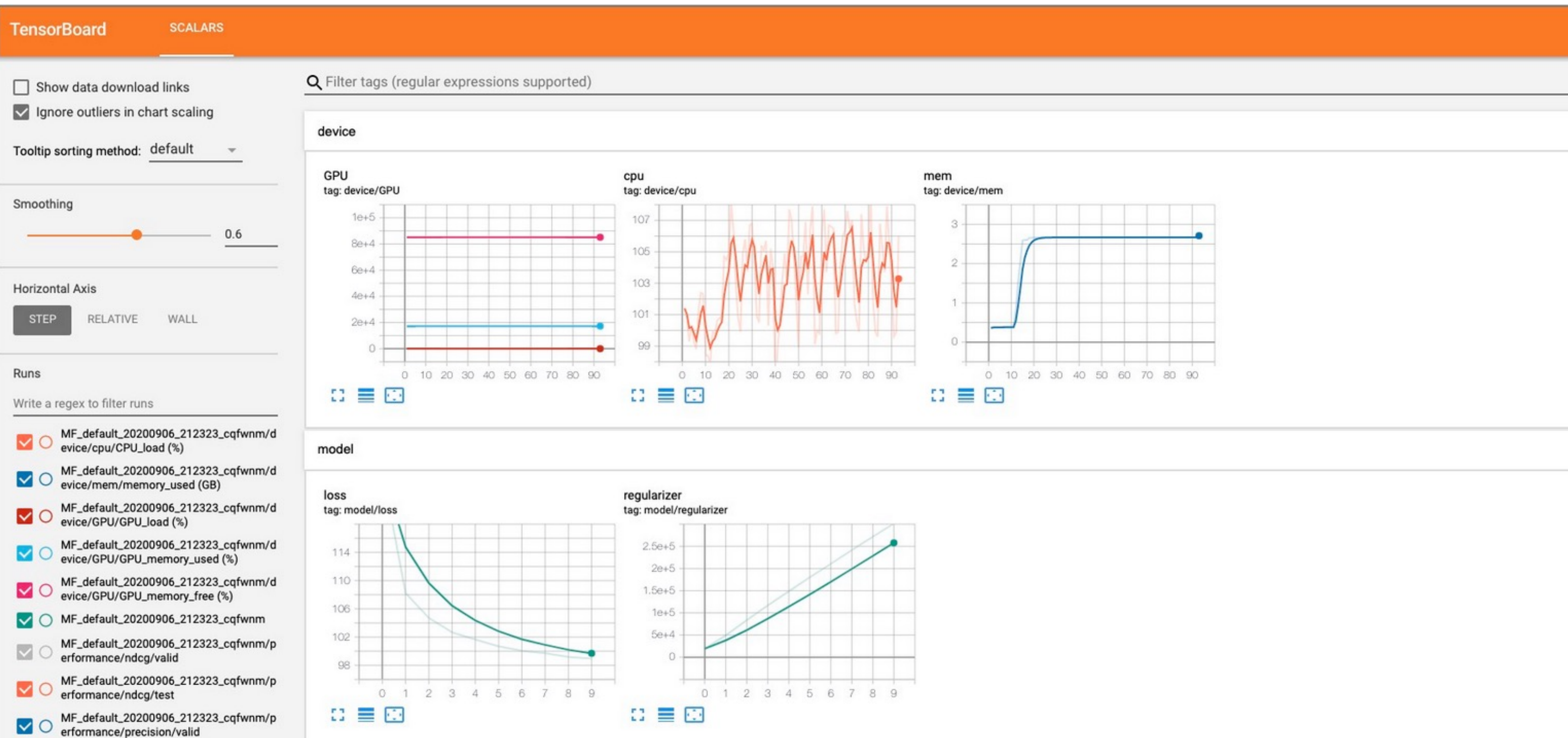
Built-in Models

- MF**: Matrix Factorization.
 - NCF**: Neural Collaborative Filtering.
 - NGCF**: Neural Graph Collaborative Filtering.
 - LightGCN**: Simplifying and Powering Graph Convolution Network for Recommendation.
 - CMN**: Collaborative Memory Network.
 - Triple2Vec**: Representing and Recommending Shopping Baskets with Complementarity, Compatibility, and Loyalty.
 - SasRec**: Self-attentive sequential recommendation.

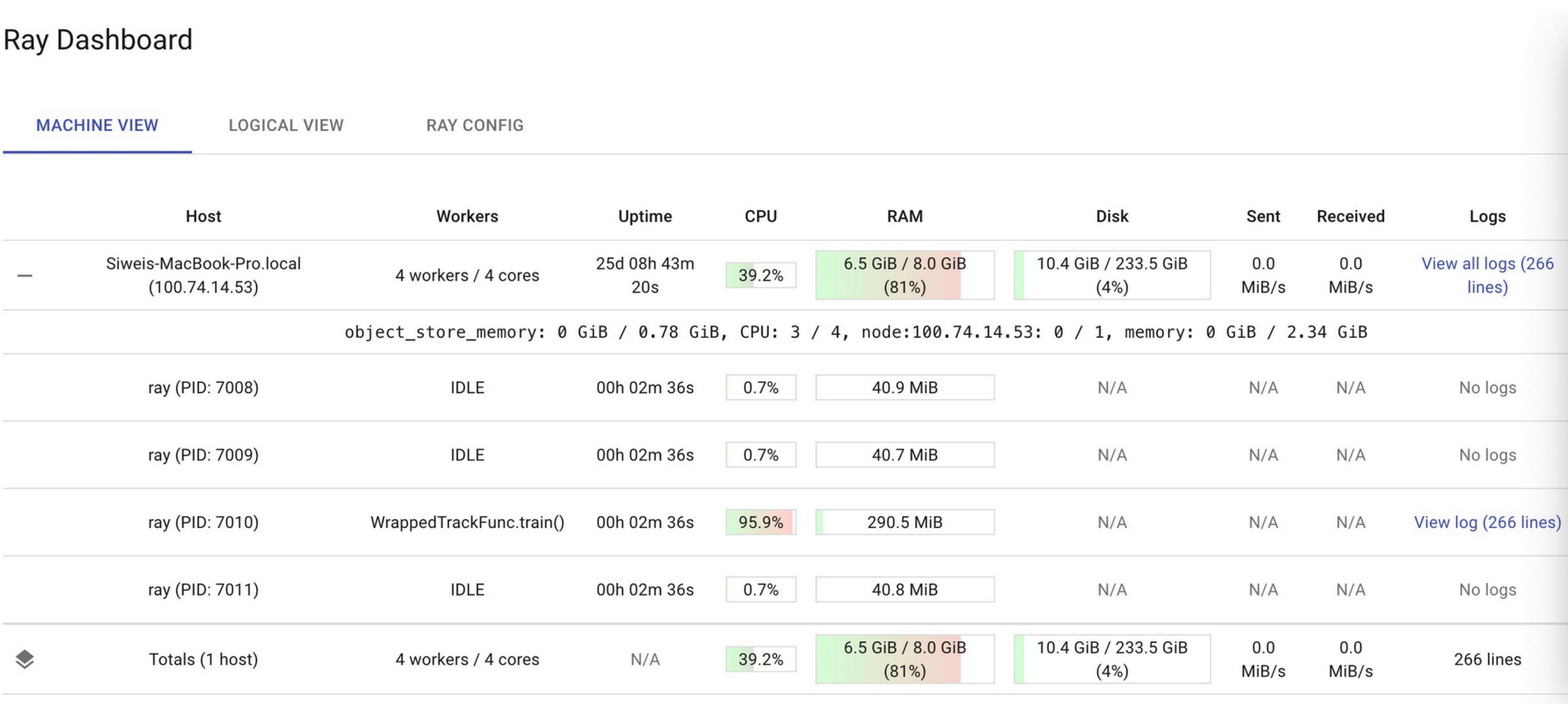
Native Dataset Support

Dataset	Interactions	Baskets	Temporal
MovieLens-100K	✓	✗	✓
MovieLens-1M	✓	✗	✓
MovieLens-25M	✓	✗	✓
Last.FM	✓	✗	✗
Epinions	✓	✗	✗
Tafeng	✓	✗	✓
Dunhumby	✓	✓	✓
Instacart	✓	✗	✓
citeulike-a	✓	✗	✗
citeulike-t	✓	✗	✗
HetRec MoiveLens	✓	✗	✓
HetRec Delicious	✓	✓	✗
HetRec LastFM	✓	✓	✓
Yelp	✓	✗	✓
Gowalla	✓	✗	✓
Yoochoose	✓	✗	✓
Diginetica	✓	✗	✓
Taobao	✓	✗	✓
Ali-mobile	✓	✗	✓
Retailrocket	✓	✗	✓

Tensorboard Integration



Ray Hyperparameter Tuning



Download Now

- Source Code**
 - <https://github.com/beta-team/beta-recsys>
 - Documentation**
 - <https://beta-recsys.readthedocs.io/>



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